

<i>Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.</i>						
Team Members	Role/ Responsibility	Email				
Alieu O Jobe	Data prep lead: loading, cleaning, imputation, features, standardization, optimization benchmark, GitHub repo and README	a.job@alustudent.com				
Dianah Gasasira Shimwa	PCA analysis lead: covariance, eigendecomposition, sorting, projection, plots and written answers	d.shimwa@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Download the OWID energy data, load it with numpy, keep 57 African countries (2000 to 2022) and add region	Alieu O Jobe	27 Sep 2026	Completed	Yes	1,299 rows kept, numpy only
	Convert columns to numbers, compute GDP per person, impute missing values (country, then region, then overall median)	Alieu O Jobe	27 Sep 2026	Completed	Yes	240 missing cells filled, 0 left
	Build the 16 features: log transforms and one hot encoding of region	Alieu O Jobe	28 Sep 2026	Completed	Yes	Region one hot encoded so no fake order
	Standardize the data with $z = (x - \text{mean}) / \text{std}$ (Step 1)	Alieu O Jobe	28 Sep 2026	Completed	Yes	Means 0 and std 1 checked
	Covariance matrix (Step 3) and the written answer on why we need it	Dianah Gasasira Shimwa	28 Sep 2026	Completed	Yes	16 by 16, symmetric confirmed
	Eigendecomposition and sorting components by explained variance (Steps 4 and 5)	Dianah Gasasira Shimwa	29 Sep 2026	Completed	Yes	Used eigh because the matrix is symmetric

	Choose the number of components dynamically, project the data and output the reduced data (Steps 6 and 7)	Dianah Gasasira Shimwa	29 Sep 2026	Completed	Yes	90% threshold gave 8 components, 93.57% variance kept
	Before and after PCA plots, scree plot and the three written interpretations (Step 8)	Dianah Gasasira Shimwa	29 Sep 2026	Completed	Yes	Each answer kept to 5 lines
	Task 3 optimization benchmark, README, GitHub repo and this task sheet	Alieu O Jobe	29 Sep 2026	Completed	Yes	Tested up to 1 million rows

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
28 Sep 2026		Alieu O Jobe	Alieu O Jobe, Dianah Gasasira Shimwa	Chose the OWID energy data for African countries. Agreed who takes data prep and who takes the PCA steps.	Alieu starts loading and cleaning. Dianah reads up on eigendecomposition.	
29 Sep 2026		Dianah Gasasira Shimwa	Alieu O Jobe, Dianah Gasasira Shimwa	Reviewed the imputation choices and the log transforms. Confirmed the standardized data has mean 0 and std 1.	Dianah finishes the eigen steps and plots. Alieu starts the optimization benchmark.	
30 Sep 2026		Alieu O Jobe	Alieu O Jobe, Dianah Gasasira Shimwa	Checked every output, the written answers and the benchmark results. Agreed on the final README wording.	Pushes the final version to GitHub and submits.	